

Reinforcement Learning Report

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Introduction

Reinforcement learning (RL) concerns itself with how agents tend to take actions in an environment to maximize net reward. A challenge in this domain is the exploration-exploitation dilemma: an agent must choose between exploiting actions known to yield high rewards and exploring alternative actions that might lead to even better outcomes. This tension pervades many real-world applications, from clinical trials and online advertising to robotics and game playing.

Chapter 1

The Exploration-Exploitation Dilemma

1.1 Problem Formulation

Consider the k -armed bandit problem for understanding the exploration-exploitation dilemma. An agent faces k different actions (arms), each with an unknown expected reward. At each time step t , the agent selects an action A_t and receives a reward R_t drawn from a probability distribution associated with that action.

Let $q_*(a)$ denote the true value of action a , defined as:

$$q_*(a) = \mathbb{E}[R_t \mid A_t = a] \quad (1.1)$$

The agent maintains estimates $Q_t(a)$ of these true values and must decide which action to select at each time step. The fundamental dilemma arises because:

- **Exploitation:** Selecting $A_t = \arg \max_a Q_t(a)$ maximizes immediate expected reward based on current knowledge
- **Exploration:** Selecting actions with uncertain values may reveal that they are actually superior to the current best estimate

1.2 The Cost of Ignorance

Pure exploitation can lead to suboptimal long-term performance. If the agent always selects the action with the highest current estimate, it may never discover that another action has a truly higher value. This is particularly problematic when initial estimates are inaccurate.

Conversely, excessive exploration wastes opportunities to earn high rewards. Random action selection, while ensuring all actions are tried, forgoes the benefits of accumulated knowledge. The challenge is to explore intelligently—focusing on actions that have genuine potential to be optimal.

1.3 Regret as a Performance Metric

The quality of an exploration strategy can be quantified through the concept of regret. Regret measures the cumulative difference between the rewards that would have been obtained by always selecting the optimal action and the rewards actually obtained. Effective exploration strategies aim to minimize regret growth.

Chapter 2

Upper Confidence Bound (UCB) Algorithm

2.1 Principle of Optimism Under Uncertainty

The Upper Confidence Bound algorithm addresses the exploration-exploitation dilemma through the principle of optimism under uncertainty. Rather than using point estimates of action values, UCB maintains confidence intervals and selects actions based on their upper confidence bounds. This approach naturally balances exploration and exploitation: actions with high uncertainty have wide confidence intervals, making their upper bounds high, while actions with high estimated values also have high upper bounds.

2.2 Mathematical Formulation

The UCB1 algorithm, introduced by Auer, Cesa-Bianchi, and Fischer (2002), selects actions according to:

$$A_t = \arg \max_a \left[Q_t(a) + c \sqrt{\frac{\ln t}{N_t(a)}} \right] \quad (2.1)$$

- $Q_t(a)$ is the estimated value of action a at time t
- $N_t(a)$ is the number of times action a has been selected prior to time t
- $c > 0$ is a parameter controlling the degree of exploration

The exploration term $c \sqrt{\frac{\ln t}{N_t(a)}}$ serves as a bonus that decreases as an action is selected more frequently. This bonus captures the uncertainty in the estimate: actions that have been tried less often receive a larger bonus.

The algorithm's simplicity and theoretical foundation make it attractive for many applications. However, it assumes reward distributions remain stationary and may not adapt quickly in non-stationary environments without modification.

Chapter 3

Optimistic Initialization Strategy

3.1 Motivation and Concept

Optimistic initialization is an elegant technique that encourages exploration through careful initialization of action-value estimates. By setting initial estimates $Q_1(a)$ higher than the true expected values, the algorithm creates an incentive to explore all actions early in learning. This method is particularly effective because it requires no changes to the action selection rule—exploration emerges naturally from optimistic beliefs that must be corrected through experience.

3.2 Implementation

Consider a greedy policy that always selects $A_t = \arg \max_a Q_t(a)$. With realistic initial values (e.g., $Q_1(a) = 0$ for all a), this policy may never explore beyond the first action that provides positive reward. With optimistic initialization, we set:

$$Q_1(a) = Q_{\text{opt}} \quad \forall a \tag{3.1}$$

where Q_{opt} is chosen to be larger than any plausible true action value. For example, if rewards are bounded in $[0, 1]$, we might set $Q_{\text{opt}} = 5$.

When the greedy policy selects an action and receives a reward below Q_{opt} , the estimate decreases. This makes other actions (still at their optimistic initial value) appear more attractive, encouraging the agent to try them. Over time, all actions are explored, and estimates converge toward true values.

3.3 Relationship to Prior Knowledge

Optimistic initialization can be viewed as encoding prior beliefs about action values. In Bayesian terms, it represents an optimistic prior distribution. This perspective connects to broader Bayesian approaches in reinforcement learning, where uncertainty drives exploration through mechanisms like Thompson sampling or Bayesian UCB.

Chapter 4

Incremental Update Methods

4.1 Introduction

Incremental update methods are a class of algorithms that allow an agent to update its understanding of the environment and optimal actions with each new experience, rather than waiting until an entire sequence of events is complete. This makes them highly suitable for online, real-time learning scenarios.

4.2 Core Concepts

The central idea behind incremental updates is to continuously refine the agent's **value function** or **policy** as new data arrives. This is achieved by adjusting the current estimates based on the difference between the expected value and the observed outcome, often called the prediction error.

Temporal Difference learning is a key incremental approach that learns directly from raw experience without a model of the environment's dynamics.

This fundamental algorithm updates the value estimate for a state based on the immediate reward and the estimated value of the next state. It uses the "value of the next state" as a proxy for the final outcome, a process known as "bootstrapping." The updates occur with every time step.

4.3 Popular Algorithms

4.3.1 Q-Learning

Q-learning is a widely used TD control algorithm. It learns the optimal policy by estimating the optimal action-value function, $Q(s, a)$, which represents the expected future rewards for taking action a in state s .

The Q-value is updated incrementally after each action. A key feature is that it uses the maximum possible Q-value from the next state to guide the learning process, effectively looking one step ahead and choosing the best perceived option for the future.

4.3.2 SARSA (State-Action-Reward-State-Action)

SARSA is another TD control algorithm. Unlike Q-learning, which uses the maximum future Q-value, SARSA updates the current action's value based on the specific action the agent actually takes in the next state, following its current policy. This makes it more sensitive to the actual path taken by the agent.

- **Optimistic initialization + UCB:** Optimistic initialization ensures early exploration of all actions, while UCB provides principled exploration throughout learning with theoretical guarantees.
- **Incremental updates + UCB:** Efficient incremental updates enable real-time computation of action values needed by UCB, with constant step-sizes allowing adaptation in non-stationary environments.
- **Optimistic initialization + incremental updates:** Optimistic initial values combined with appropriate step-sizes create early exploration pressure that gradually diminishes as estimates improve.